🎨 Image Cartoonification Using Computer Vision

Abstract

This project focuses on converting normal images into cartoon-style images using computer vision techniques. The system uses **filtering, edge detection, and color quantization** to transform photos into visually appealing cartoons. An interactive **Streamlit dashboard** allows users to upload images, select cartoon styles, and adjust edge and smoothing parameters in real-time. This project demonstrates the application of image processing, machine learning concepts, and interactive dashboards in computer vision.

Introduction

With the rise of social media and digital art, cartoonification of images has become a popular trend. Traditional cartoon effects are often manual and time-consuming. Using **computer vision techniques**, images can be automatically converted into cartoons by detecting edges, smoothing colors, and reducing the number of color levels.

This project creates an interactive tool that allows users to convert images into multiple cartoon styles. The system uses **OpenCV** for image processing and **Streamlit** for the interactive interface. Users can see real-time previews and download the processed images.

Objective

The main objectives of this project are:

1. To convert normal images into cartoon-style images using image processing techniques.
2. To provide multiple cartoon styles (Classic Cartoon, Sketch, Posterized).
3. To allow interactive adjustment of edge strength and smoothing parameters.
4. To create a user-friendly web interface for image upload, processing, and download.
5. To demonstrate the application of computer vision and Python programming for creative projects.

Tools and Technologies

Tool / Library	Purpose
Python	Programming language
OpenCV	Image processing (edge detection, filtering)
NumPy	Array and matrix operations

Tool / Library	Purpose
Pillow (PIL)	Image loading and saving
Streamlit	Interactive web application

Methodology

Step 1: Image Upload

- Users upload images in JPG, JPEG, or PNG format.

Step 2: Style Selection

- Dropdown menu allows selecting a cartoon style:
    - Classic Cartoon
    - Sketch / Pencil Drawing
    - Posterized / Pop Art

Step 3: Image Processing

1. **Classic Cartoon:**
    - Apply **bilateral filter** to smooth colors while preserving edges.
    - Detect edges using **Canny edge detection**.
    - Combine edges with the smoothed image.
2. **Sketch / Pencil:**
    - Convert image to grayscale.
    - Invert and blur the grayscale image.
    - Combine to create pencil-drawing effect.
3. **Posterized / Pop Art:**
    - Reduce number of colors using **K-means clustering**.
    - Apply smoothing and combine with edges.

Step 4: Interactive Adjustments

- Users can control:
    - **Edge Strength** – Adjust intensity of edges.
    - **Smoothing Level** – Adjust color smoothing.
    - **Color Levels** – Adjust only for Posterized style.

Step 5: Output and Download

- The processed cartoon image is displayed alongside the original.
- Users can download the final cartoon image.

Results

- Images are successfully converted to cartoon styles.
- Users can select styles and adjust parameters interactively.
- Real-time preview allows immediate feedback on adjustments.
- Download option enables saving the final cartoon image.

Example Outputs:

- Classic Cartoon – Smooth colors with clear edges.
- Sketch – Pencil-drawing effect in grayscale.
- Posterized – Reduced colors with bold edges.

Applications

- Social media photo filters and apps
- Digital art and creative projects
- Educational demonstration of computer vision
- Fun interactive tools for personal use

Advantages

- Interactive and user-friendly interface
- Multiple cartoon styles for creativity
- Real-time parameter adjustment
- Lightweight and fast processing

Limitations

- Works best with clear, high-quality images
- Posterized style may lose details with very low color levels
- Not real-time for video or webcam input (current version)

Conclusion

This project demonstrates how **computer vision techniques** such as filtering, edge detection, and color quantization can be applied to create cartoon images. With **Streamlit**, the system provides an interactive and user-friendly interface for image processing. The project successfully converts images into multiple cartoon styles with adjustable parameters and download options. This project highlights the application of Python programming and computer vision in creative digital media projects.